

SUDARSHAN UAV — OPERATOR CERTIFICATION QUIZ

25-Question Multiple Choice | Time Allowed: 40 Minutes

Name: _____ Date: _____ Score: _____ / 25

Instructions: Circle the single best answer for each question. No open-book unless directed by instructor.

Section 1 — Hardware & Wiring

Q1. Which I2C address does MPU6050 use when AD0 = LOW?

- (a) 0x40
- (b) 0x68 ✓
- (c) 0x69
- (d) 0x76

Q2. What does the Mega → ESP32 voltage divider accomplish?

- (a) Boost 3.3 V → 5 V
- (b) Convert 5 V → 3.3 V ✓
- (c) Filter noise
- (d) Regulate power

Q3. Which PCA9685 channel drives the Front-Right motor?

- (a) CH0
- (b) CH1 ✓
- (c) CH2
- (d) CH3

Q4. Which Uno pins connect to the PCA9685 I2C bus?

- (a) D2/D3
- (b) D18/D19
- (c) A4/A5 ✓
- (d) D6/D7

Q5. Why must Uno Pins 0 and 1 be disconnected before USB flashing?

- (a) They carry too much current
- (b) They are shared with the USB-serial adapter ✓
- (c) They can damage the PCA9685
- (d) They interfere with I2C

Section 2 — Flight Controller

Q6. What is the Mega FC control-loop rate?

- (a) 50 Hz
- (b) 100 Hz
- (c) 200 Hz
- (d) 250 Hz ✓

Q7. CF_ALPHA = 0.98 means the complementary filter applies:

- (a) 98% accelerometer, 2% gyroscope
- (b) 98% gyroscope, 2% accelerometer ✓
- (c) 98% sonar
- (d) It is a sampling ratio

Q8. The drone spins CW continuously with no yaw command. What is the correct fix?

- (a) Increase Ki_yaw
- (b) Swap all propellers
- (c) Change IMU_YAW_SIGN to -1 ✓
- (d) Decrease CF_ALPHA

Q9. What happens when sonar reads < 8 cm for 6 consecutive counts during LAND?

- (a) Drone disarms immediately ✓
- (b) Drone hovers at ground level
- (c) Slow hover begins at 30 cm
- (d) LAND mode continues

Q10. Which command is BLOCKED while the drone is armed?

- (a) HOVER
- (b) OVERRIDE
- (c) LAND
- (d) SET_MOTOR_MAP ✓

Section 3 — Communications

Q11. What is the motor packet format from Mega → Uno?

- (a) JSON over UART
- (b) 10-byte binary + XOR checksum ✓
- (c) 8-byte CAN frame
- (d) PWM pulses

Q12. Which TCP port does the Python GCS use to connect to the ESP32?

- (a) 80
- (b) 81
- (c) 5760 ✓
- (d) 8080

Q13. At what rate does the FC send telemetry to the GCS?

- (a) 1 Hz
- (b) 5 Hz
- (c) 10 Hz ✓
- (d) 25 Hz

Q14. Where must the AES encryption key be stored?

- (a) In the git repository
- (b) In credentials.h / credentials.py only, never committed ✓
- (c) In the project PDF
- (d) Sent inside every TCP packet

Q15. The XOR checksum covers which bytes of the 10-byte motor packet?

- (a) All 10 bytes
- (b) Bytes 0–8
- (c) Bytes 1–8 ✓
- (d) Bytes 1–9

Section 4 — Safety & Emergency

Q16. How many independent DMS (Dead-Man Switch) layers are implemented?

- (a) 1
- (b) 2
- (c) 3 ✓
- (d) 4

Q17. What is the FC-side DMS timeout?

- (a) 10 s
- (b) 20 s
- (c) 30 s ✓
- (d) 60 s

Q18. A battery voltage of 9.9 V triggers which action?

- (a) Warning flag only
- (b) Telemetry flag only
- (c) Automatic LAND command ✓
- (d) Immediate motor cut

Q19. After a KILL command is executed, to resume flying you must:

- (a) Send ARM
- (b) Send DISARM then ARM
- (c) Power-cycle the drone ✓
- (d) Press HOVER

Q20. sonar_ok = 0 is reported during a hover. The correct operator action is:

- (a) Continue normally
- (b) Increase altitude slowly
- (c) Land manually immediately ✓
- (d) Switch to OVERRIDE mode

Section 5 — Operations

Q21. What is the correct power-on sequence?

- (a) ESP32 first
- (b) Mega + Uno first, wait for ESC beeps, then ESP32 ✓
- (c) All components simultaneously
- (d) Phone / GCS first

Q22. What must be done before executing MOTOR_TEST?

- (a) ARM the drone
- (b) Remove all propellers ✓
- (c) Connect phone GPS
- (d) Enable AES encryption

Q23. What is the Web GCS session priority override code?

- (a) 0000
- (b) 980752
- (c) 1410 ✓
- (d) admin

Q24. What is the minimum safety perimeter before executing ARM outdoors?

- (a) 1 m
- (b) 5 m
- (c) 10 m ✓
- (d) 20 m

Q25. PRESET mission segments use which positional reference?

- (a) GPS coordinates
- (b) Bearing + distance relative to IMU heading ✓
- (c) X/Y grid coordinates
- (d) Compass waypoints

ANSWER KEY — INSTRUCTOR COPY

Do Not Distribute to Candidates

No.	Answer	No.	Answer	No.	Answer	No.	Answer	No.	Answer
1	B	2	B	3	B	4	C	5	B
6	D	7	B	8	C	9	A	10	D
11	B	12	C	13	C	14	B	15	C
16	C	17	C	18	C	19	C	20	C
21	B	22	B	23	C	24	C	25	B

Scoring Guide

Score Range	Status	Notes
23 – 25	CERTIFIED OPERATOR	May operate SUDARSHAN UAV independently
18 – 22	SUPERVISED ONLY	Must fly under certified operator supervision
< 18	REQUIRES RETRAINING	Must complete refresher training before re-assessment